

Numerical Analysis I

Homework 4

Exercises:

1. EX.3.1.1.b2c [handwritten](#)
2. EX.3.1.2.b2c [handwritten](#)
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EX.3.1.1.b2c, Sauer3

Use Lagrange interpolation to find a polynomial that passes through the points

b. $(-1, 0), (2, 1), (3, 1), (5, 2)$.

c. $(0, -2), (2, 1), (4, 4)$.

EX.3.1.2.b2c, Sauer3

Use Newton's divided differences to find the interpolating polynomials of the points in EX.3.1.1.b2c, and verify agreement with the Lagrange interpolating polynomial.

b. $(-1, 0)$, $(2, 1)$, $(3, 1)$, $(5, 2)$.

c. $(0, -2)$, $(2, 1)$, $(4, 4)$.

EX.3.1.4, Sauer3

- a. Find a polynomial $p(x)$ of degree 3 or less whose graph passes through the points $(0, 0)$, $(1, 1)$, $(2, 2)$, $(3, 7)$.
- b. Find two other polynomials (of any degree) that pass through these four points.
- c. Decide whether there exists a polynomial $p(x)$ of degree 3 or less whose graph passes through the points $(0, 0)$, $(1, 1)$, $(2, 2)$, $(3, 7)$, and $(4, 2)$.

EX.3.1.6, Sauer3

Write down a polynomial of degree exactly 5 that interpolates the four points $(1, 1)$, $(2, 3)$, $(3, 3)$, $(4, 4)$.

EX.3.1.8, Sauer3

Let $p(x)$ be the degree 9 polynomial that takes the value 112 at $x = 1$, takes the value 2 at $x = 10$, and equals zero for $x = 2, \dots, 9$. Calculate $p(0)$.

EX.3.1.10, Sauer3

Let $p(x)$ be the degree 5 polynomial that takes the value 10 at $x = 1, 2, 3, 4, 5$ and the value 15 at $x = 6$. Find $p(7)$.

EX.3.1.18, Sauer3

The expected lifetime of an industrial fan when operated at the listed temperature is shown in the table that follows. Estimate the lifetime at 70°C by using (a) the parabola from the last three data points (b) the degree-3 curve using all four points.

temp (°C)	hrs ($\times 1000$)
25	95
40	75
50	63
60	54

Please use COLAB for this problem.

No handwritten work! Please turn in a Colab Notebook

CP.3.1.4, Sauer3

Remodel the `sin1` calculator key in Program 3.3 to build `cos1`, a cosine key that follows the same principles. First decide on the fundamental domain for cosine.

EX.3.2.2, Sauer3

- a. Given the data points $(1, 0)$, $(2, \ln(2))$, and $(4, \ln(4))$, find the degree 2 interpolating polynomial.
- b. Use the result of (a) to approximate $\ln(3)$.
- c. Use Theorem 3.3 to give an error bound for the approximation in part (b).
- d. Compare the actual error to your error bound.